t3-w05 — condition C2

Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

> C1 plus the lesson note as source evidence.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	**present**

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

- identify properties of cylinders and spheres
- find the volume of a cube and a cuboid

entry_behaviour

- properties of cubes and cuboid
- properties of pyramids and cones

instructional_materials

- shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

objective_text

identify properties of cylinders and spheres

current_step

1

instructional_events

- present_content
- elicit_performance
- present_content

student_tasks

- 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

source_figures


```
[
 {
  "caption": "A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "A sphere drawn as a circle with an ellipse across its middle to show the curved surface.",
  "figure_ref": "fig-02"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Properties of a cylinder\n\n**fig-01.** A cylinder drawn upright: an ellipse for the top face, two vertical sides\nfalling from its ends, and a curve closing the base.\n\n[drawing attached to this section as fig-01]\n\n1. It has two circular faces and one curved face\n2. It has no vertex\n3. It has two curved edges\n4. It has three faces altogether\n\n#### Properties of sphere\n\n**fig-02.** A sphere drawn as a circle with an ellipse across its middle to show the\ncurved surface.\n\n[drawing attached to this section as fig-02]\n\n1. A sphere is symmetrical from all direction\n2. It has no edge and no vertex\n3. It has one curved surface\n\n#### Student activity\n\n1. How many circular faces does a solid cylinder have?\n2. A cylinder has how many faces."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — find the volume of a cube and a cuboid

objective_text

find the volume of a cube and a cuboid

current_step

2

instructional_events

• present_content
• provide_learning_guidance

assessment_points

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

source_figures


```
[
 {
  "caption": "A triangular prism lying on its rectangular face. The triangular end has a base of
5 m and a height of 3 m, and the prism runs back 10 m.",
  "figure_ref": "fig-03"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Volume of cube and cuboid\n\n1. Volume of cuboid = Area of base $\\times$
height\n\n   $= \\text{length} \\times \\text{breadth} \\times \\text{height}$\n\n   $= lbh$
\n\n2. Volume of cube = Length $\\times$ Length $\\times$ Length\n\n   $= l^{3}$\n\n**Example
1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 3cm$\n\nbreadth $= 2cm$\n\nheight $=
8cm$\n\nVolume $= 3cm \\times 2cm \\times 8cm$\n\n$= 48cm^{3}$\n\n**Example 2.** The volume of a
cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the\nheight of the cuboid.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 6cm$\n\nbreadth $= 2cm$\n\nheight
$= ?$\n\nVolume $= 24cm^{3}$\n\n$24cm^{3} = 6cm \\times 2cm \\times h$\n\n$24cm^{3} = 12cm^{2}h$
\n\nDivide both side by 12cm$^{2}$\n\n$\\frac{24cm^{3}}{12cm^{2}} = \\frac{12cm^{2}h}{12cm^{2}}$
\n\n$2cm = h$\n\n$h = 2cm$\n\n#### Volume of cube\n\n**Example 3.** Calculate the volume of a
cube of side 3cm.\n\nSolution\n\nVolume of cube $= l^{3}$\n\n$V = (3cm)^{3}$\n\n$V = 3cm \\times
3cm \\times 3cm$\n\n$V = 27cm^{3}$\n\n#### Volume of prism\n\nCalculate the volume of the prism
below.\n\n**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a
base\nof 5 m and a height of 3 m, and the prism runs back 10 m.\n\n[drawing attached to this
section as fig-03, marked 5 m, 3 m, 10 m]\n\nSolution:\n\nArea of triangular face $= \\frac{1}{2}
bh$\n\n$= \\frac{1}{2} \\times 5 \\times 3$\n\n$= 7.5m^{2}$\n\nVolume of prism $=$ Area of base $
\\times$ height\n\n$= 7.5m^{2} \\times 10m$\n\n$= 75m^{3}$"
 }
]
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Properties of Cylinders and Spheres

EXPLANATION

A cylinder is a solid with two identical circular faces (the top and bottom) and one curved side that connects them.
- Circular faces: 2
- Curved face: 1
- Total faces: 3
- No vertices (points where edges meet)
- No straight edges; the side is a curved surface.

A sphere is a perfectly round solid where every point on the surface is the same distance from the centre.
- It has no faces, edges, or vertices in the usual sense.
- It is completely symmetrical from every direction.
- Its surface is a single continuous curved surface.

These properties help you recognise each shape and describe them correctly.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
  \draw (-1.2,2.6) -- (-1.2,0);
  \draw (1.2,2.6) -- (1.2,0);
  \draw (-1.2,0) arc (180:360:1.2 and 0.35);
  \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) circle (1.3);
  \draw (-1.3,0) arc (180:360:1.3 and 0.42);
  \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
\end{tikzpicture}
```

1. 1. How many circular faces does a solid cylinder have?
answer: 2

2. 2. A cylinder has how many faces in total?
answer: 3

Objective 2

Finding the Volume of Cubes and Cuboids

EXPLANATION

The volume of a solid tells us how much space it occupies.

Cuboid

A cuboid has three dimensions: length (l), breadth (b) and height (h). Its volume is found by multiplying these three measurements:

$$V_{\text{cuboid}} = l \times b \times h$$

Cube

A cube is a special cuboid where all three edges are the same length (s). Its volume is therefore:

$$V_{\text{cube}} = s \times s \times s = s^{3}$$

Remember to keep the units consistent (e.g., all in cm or all in m) and write the final answer with the unit cm^3 or m^3 as appropriate.

WORKED_EXAMPLE

Example 1 – Volume of a cuboid

Find the volume of a cuboid whose length is 3 cm, breadth is 2 cm and height is 8 cm.

Solution:

$$V = l \times b \times h = 3\,\text{cm} \times 2\,\text{cm} \times 8\,\text{cm}$$
$$V = 48\,\text{cm}^3$$

Example 2 – Finding the missing height of a cuboid

The volume of a cuboid is 24 cm^3. Its base measures 6 cm by 2 cm. What is its height?

Solution:

$$V = l \times b \times h$$
$$24 = 6 \times 2 \times h$$
$$24 = 12 \times h$$
$$h = \frac{24}{12} = 2\,\text{cm}$$

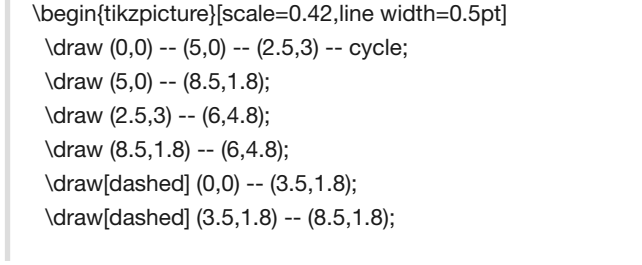
Example 3 – Volume of a cube

Calculate the volume of a cube with side length 3 cm.

Solution:

$$V = s^{3} = (3\,\text{cm})^{3} = 3\,\text{cm} \times 3\,\text{cm} \times 3\,\text{cm} = 27\,\text{cm}^3$$

VISUAL



```
  \draw[dashed] (3.5,1.8) -- (6,4.8);
  \draw[dashed] (2.5,0) -- (2.5,3);
  \node[below] at (2.5,-0.15) {$5\,\mathrm{m}$};
  \node[left] at (2.35,1.5) {$3\,\mathrm{m}$};
  \node[below right] at (6.9,0.8) {$10\,\mathrm{m}$};
\end{tikzpicture}
```

1. 1. Calculate the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.
answer: 96 cm^3

2. 2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
answer: 30 m^3

3. 3. What is the volume of a cube that is 4 cm × 4 cm × 4 cm?
answer: 64 cm^3

4. 4. A rectangular tank 6 m long by 2 m wide holds 36 m^3 of water. How deep is the water in the tank?
answer: 3 m

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Enacted content	**grounded**
Worked example a problem with its solution worked through	1	Enacted content	**grounded**
Figure a drawing the learner reads from	3	Figures	**grounded**
Practice what the learner does	6	Set work	**grounded**
Answer what the learner checks against	6	Enacted content	**grounded**